Fatima Laraib

fatimalaraib06@gmail.com | 03365633017 | Islamabad | LinkedIn | GitHub | Portfolio

Summary

Data Science student (GPA 3.98/4) with hands-on experience in Python, SQL, machine learning, NLP and signal processing. Experienced in building end-to-end projects, from neural networks and computer vision models to LLM-integrated and full-stack applications.

Education

FAST NUCES **May 2028**
Bachelors in Data Science *Islamabad*
- GPA: 3.98/4
- Gold Medalist x4
- Dean's and Rector's listed

IMCG I-10/4 **2022-2024**
FSc Pre-Engineering *Islamabad*
- Aggregate: 92%

Experience

XCORRE Internship **July 2026-October 2026**
AI Engineer *Remote*
- Built the backend for a web-application using FastAPI
- Implemented fingerprint scanner and camera integration to record patient data in the database and identify patients
- Built a disease predicting model for chest X-rays using TorchXRayVision and DenseNet
- Built and managed PostgreSQL and Supabase databases for the web-application

KDD Internship **July 2026**
Data Analysis and Visualization *Islamabad*
- Extracted and cleaned data from online sources
- Used NLPs and NLIs to build confidence scoring models
- Used d3.js to build interactive visualizations

Teaching Assistant, FAST NUCES **Sept 2026-Present**
Linear Algebra *Islamabad*
- Assist in teaching Linear Algebra and grade quizzes for undergraduate students

Lab Demonstrator, FAST NUCES **Jan 2026-May 2026**
Experimental Design and Quantitative Analysis *Islamabad*
- Delivered hands-on lab instruction to undergraduates in experimental design, data collection and quantitative analysis
- Graded technical lab reports for 40+ students per cohort, assessing data interpretation and statistical reasoning
- Explained complex technical concepts in accessible terms, improving student comprehension

Marketing Intern, E Deviser **Sept 2025-Oct 2025**
Content and Analytics *Islamabad*
- Monitored content engagement metrics to identify trends and inform content strategy
- Produced data-driven creative briefs and visual assets aligned with audience analytics

Projects

Vocal-to-Piano Melody Transcription
- Built an audio pipeline in Python (librosa, SciPy) to extract pitch from vocal recordings using FFT and autocorrelation
- Converted pitch data into structured MIDI output, producing piano melodies from raw singing

Flappy Bird Neural Network Simulator
- Built a neural network from scratch in Python
- Flappy Bird Simulation built using Pygame
- Designed a Markov chain-based procedural maze generator, validated with traversal algorithms to ensure solvable paths

AI-powered Chatbot
- Built a chatbot using OpenAI API key and Python, with context management, prompt engineering and error handling for multi-turn dialogue
- Object detection using YOLOv8 for voice chat

Study Buddy Networking App
- Developed a full-stack web application (Python/JavaScript) enabling students to connect by subject interests and availability
- Managed user data storage and retrieval with a MySQL/PostgreSQL backend

RideFlow
- Built a full-stack ride-hailing web application with role-based access for Riders, Drivers and Admins
- Riders book and pay for rides, rate drivers and track spending; drivers register vehicles, accept requests and view earnings and payouts
- Admins verify drivers and vehicles, view earnings dashboards, ban or suspend accounts and set the base fare
- Tech stack: MySQL, Express/Node.js, HTML/CSS

Movie Recommender
- Built a genre-based movie recommendation system in C++ over 5,000 movie records
- Implemented trees, hash tables and graphs to store, search and relate movies

Skills and Certificates

- **Technologies:** Python, C++, JavaScript, TypeScript, SQL, Pandas, NumPy, scikit-learn, PyTorch, React, D3.js, Node.js, Express, Flask, FastAPI, MySQL, PostgreSQL, Supabase, MongoDB, Firebase, Pinecone, OpenAI API, Git, GitHub, Power BI, REST APIs
- **Skills:** Machine Learning, Fine-tuning, NLPs, NLIs, Neural Networks, Probabilistic Modelling, Signal Processing, Data Wrangling
- **Certificates:** What is Data Science? (IBM - Coursera); Essentials of Prompt Engineering (AWS); Robotics Certificate (FAST NUCES); Machine Learning Specialization (IBM, ongoing)